

Grade Eight Science
Lab Discussion: What are Variables in an Experiment?

There are three types of variables in science experiments:

- Control Variable(s): Things that are kept the same in all trials of the experiment (all experiments have multiple controls)
- Dependent Variable: What you are measuring in the experiment (the data that you are collecting)
- Independent Variable: What you are changing to get the measurements (the difference between the experimental and control groups)

EXAMPLE:

Research Question: *The Pickle ball vs. The Tennis ball: Which ball will yield the greatest bounce height?*

Background Information	Describe the Background Information
Validity of the Research Question	The purpose of this experiment is to determine which type of ball, a tennis ball or a pickle ball, will bounce the highest when dropped under the same conditions. Tennis and pickleball are both racquet sports that use two racquets, a ball, and a net, but there are some differences, such as the type of ball used. The focus of this investigation is on comparing the bounce height of the tennis ball and the pickle ball to see which one bounces higher and why.
Hypothesis	If a tennis ball and a pickle ball are dropped from a height 1 m then the tennis ball will bounce the highest because tennis balls are made of rubber, while pickle balls are made of plastic, and rubber has more elasticity which will lead to the greatest bounce height
Explanation of the hypothesis	A tennis ball has a rubber core filled with pressurized air (Admin) while a pickle ball is a hollow, plastic ball with holes in it that is not pressurized. (Pickleball Superstore). The experiment will conduct three trials where each type of ball is dropped from a height of 1 meter on the same surface. The data collected will be analyzed, and it will be determined which ball bounces back at a greater height. Research will be conducted to determine why one ball bounces back at a greater height than the other.

Type of Variable	In the experiment the variable is:	Method of Measurement/ Method of Control
Independent	The type of ball	The two types of balls considered are a tennis ball and a pickle ball; a new identical ball will be used in each trial
Dependent	The bounce height of the ball	This will be measured using a meter stick attached to the wall and an iPhone camera in slow motion to record the motion
Control	Drop height of the ball (1 m)	During each trial the ball is dropped from a height of 1 m measured using the same meter stick
Control	Surface the ball is dropped on	During each trial the ball is dropped on the same tile floor
Control	Method of recording the ball drop	During each trial the iPhone camera was set to the same setting on slow motion and the same iPhone camera was used

Research Question 1:

Does adding dimples to a car improve the gas mileage?

Background Information	Describe the Background Information
Validity of the Research Question	
Hypothesis	
Explanation of the hypothesis	

Type of Variable	In the experiment the variable is:	Method of Measurement/ Method of Control
Independent		
Dependent		
Control(s)		
Control(s)		

Research Question 2:
 Are elephants afraid of mice?

Background Information	Describe the Background Information
Validity of the Research Question	
Hypothesis	
Explanation of the hypothesis	

Type of Variable	In the experiment the variable is:	Method of Measurement/ Method of Control
Independent		

Dependent		
Control(s)		
Control(s)		

MLA Works Cited:

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